

Hamza Hafeez

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Summary

Passionate AI/ML Engineer with 4+ years of experience designing and deploying scalable machine learning systems and data-driven solutions in cybersecurity and big data environments. Experienced in building Generative AI and LLM-based applications, including RAG pipelines and agentic workflows for intelligent automation and decision-making. Strong background in developing end-to-end ML/DL pipelines, from data engineering and model training to production deployment using MLOps practices. Skilled in integrating AI systems with large-scale enterprise data platforms for real-time analytics and threat detection. Focused on bridging traditional machine learning with modern AI systems to deliver impactful, production-ready solutions.

Work Experience

Sr. ML Engineer & Active Team Lead

Ebryx, Pakistan

Apr 2022 – Present

- Developed a fully agentic, multi-agent SOC automation system that autonomously performs alert triage, generates and executes investigation plans, and delivers final verdicts—reducing human analyst workload by 20%.
- Led the design and development of end-to-end ML pipelines for cybersecurity, including data analysis, feature engineering, model training, deployment, and monitoring; also architected a machine learning engine supporting traditional ML and LLM-based applications, fine-tuning open-source LLMs for domain-specific tasks.
- Built UEBA and network security models to detect anomalous user behavior and network threats using machine learning and deep learning techniques, improving threat detection and situational awareness.
- Developed and optimized AI-powered tools using LangChain and LLMs for smart document search and contextual retrieval (RAG); optimized inference performance and deployed solutions for real-world use cases.
- Designed and implemented scalable batch and real-time data pipelines for threat detection, using Apache Spark and Airflow for ETL workflows, and Kafka, Logstash, and Filebeat for streaming log ingestion and real-time analysis; integrated event-based automation with Node-RED where applicable.
- Built real-time, interactive dashboards with Streamlit and Apache Superset to visualize threats, anomalies, and data insights for operational teams.
- Deployed ML solutions in Jupyter and standalone Python environments; containerized applications using Docker to ensure reproducible and environment-consistent deployments.
- Collaborated with cybersecurity domain experts to encapsulate knowledge and generate high-quality, labeled datasets tailored for ML model training.
- Managed and mentored over 5 interns and supervised multiple undergraduate Final Year Projects focused on machine learning in cybersecurity.
- Ensured data integrity, security, and compliance across all data processing and machine learning workflows.

Research Assistant & ML Developer

IGNITE Funded National ICT & R&D Project, COMSATS University Islamabad

Dec 2018 – July 2019

- Designed and implemented a prototype testbed for hospital data management using Biomedical Wireless Sensor Networks, supporting real-time health monitoring and data acquisition.
- Developed and trained machine learning models to predict personalized medication recommendations based on patient symptoms, enabling data-driven decision support in healthcare.

Android Developer (Remote)

Talentbase

2017 – 2018

- Developed an Android-based biometric enrollment application for a pension system, enabling secure pension disbursement and accountability for retired employees.
- Integrated the DIGITAL PERSONA U4500 Fingerprint Scanner using native SDKs, enhancing accuracy and speed of biometric data capture.

Freelance Software Developer

Freelancer / Fiverr

2016 – 2020

- Delivered various client projects, including Android apps, full-stack web applications, and IoT solutions, tailored to diverse business requirements.

Technical Skills

- **Programming & Software Engineering:** Python, Java, C++, JavaScript, PHP, C#, OOP, Data Structures & Algorithms, REST APIs, Microservices Architecture
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, Keras, Supervised & Unsupervised Learning, Model Evaluation, Feature Engineering, Anomaly Detection (PyOD, DeepOD)
- **Generative AI, LLMs & Agentic Systems:** HuggingFace Transformers, vLLM, Ollama, llama-cpp-python, RAG (Retrieval-Augmented Generation), Prompt Engineering, LLM Fine-tuning (LoRA, QLoRA), LangChain, LangGraph, DeepAgents, Multi-Agent Systems, Tool Calling, NeMo Guardrails, Guardrails.ai, Microsoft Presidio (PII Redaction)
- **NLP & Language AI:** Text Classification, Named Entity Recognition (NER), Embeddings, LlamaIndex, Tokenization, Semantic Search, Transformer-based Architectures
- **Data Engineering & ETL Pipelines:** Apache Spark, Apache Airflow, Kafka, NiFi, ELK Stack (Elasticsearch, Logstash, Kibana), Event-Driven Pipelines, Workflow Orchestration
- **MLOps, DevOps & Automation:** Docker, Git, MLflow, CI/CD Pipelines, Model Deployment & Monitoring, Azure ML Notebooks, REST, GraphQL, Webhooks, Postman, n8n, Node-RED
- **Cloud Platforms:** AWS (S3, EC2, Lambda, SageMaker), Azure ML
- **Databases & Vector Stores:** MySQL, MongoDB, Milvus, ChromaDB, Elasticsearch, Firebase
- **Data Analysis & Visualization:** NumPy, Pandas, Streamlit, Grafana, Kibana, Apache Superset
- **Other Development:** Android (Java/Kotlin), Laravel, CodeIgniter, Core PHP, Arduino, ESP32, NodeMCU, BLE, IoT Integration

Projects

AI-Powered SOC Platform

- Agentic multi-agent system for autonomous alert triage, investigation planning, and verdict generation.
- **Stack:** Python, LangGraph, LangChain, OpenAI, LLMs, SIEM APIs, , RAG, Milvus, Predictive Modeling, Embeddings Models, VLLM, Docker, FastAPI

AI Chatbot for Technical Documentation

- LLM-based chatbot enabling contextual Q&A and intelligent knowledge retrieval over large-scale documentation.
- **Stack:** Python, LangChain, RAG, ChromaDB/Milvus, HuggingFace, Streamlit, Embeddings Models, Docker, FastAPI

UEBA-Based Anomaly Detection

- Behavioral analytics models detecting abnormal login, traffic, process, and working pattern anomalies.
- **Stack:** Python, PyOD, DeepOD, Scikit-learn, Kafka, Elasticsearch, Docker

Log-Based Anomaly Detection

- ML-based anomaly detection on large-scale system and network logs for security threat identification.
- **Stack:** Python, Kafka, Logstash, Scikit-learn, Elasticsearch, Docker

- *Specific project details, tools, and models are confidential due to NDA agreements.*

Multi-Source ETL Pipeline

- Scalable pipeline integrating SIEM, EDR, and log sources for real-time ingestion, processing, and analytics.
- **Stack:** Python, Kafka, Apache Airflow, Logstash, Filebeat, API Integration, Elasticsearch

DLP and EDR Security Analytics

- AI-driven detection use cases for Data Loss Prevention and Endpoint Detection & Response systems.
- **Stack:** Python, Scikit-learn, ELK Stack, REST APIs, Powershell, Windows Events

Urdu Short Text Reuse Detection (MS Thesis)

- Classification system for Urdu text similarity and short text reuse detection at sentence level.
- **Stack:** Sentence Transformers, N-Gram Embeddings, Python, Scikit-learn, Word Embeddings, Sentence Embeddings

Malware Detection System

- ML-based malware detection using dataset analysis, feature engineering, and predictive modeling.
- **Stack:** Scikit-learn, Python, Pandas, NumPy

Education

MS Computer Science, COMSATS University Islamabad, Lahore Campus (CGPA: 3.37)	2019 - 2022
BS Software Engineering, UBIT, University of Karachi (CGPA: 2.98)	2013 - 2016

Achievements

- **Academic & Industry Recognition:** Invited as a judge at SOFTEC’26 (FAST-NUCES), evaluating Final Year Projects and AI Hackathon submissions focused on Artificial Intelligence and Machine Learning.
- **Research Paper:** "Urdu Short Paraphrase Detection at Sentence Level" ACM Transactions on Asian and Low-Resource Language Information Processing, 2023. Available at: Link
- **Coding Competition:** Runner up, Web Development & Designing Competition, IBA Probattle’15, IBA Karachi, 2015.